

Md Monjurul Karim

SIAT, CAS, Shenzhen, China
+86 132 6189 0899 · karim@siat.ac.cn
Homepage

I am working at the intersection of artificial intelligence, distributed computing, and networked systems, with a focus on building trustworthy, efficient, and scalable intelligent infrastructures.

Interests: Agentic and decentralized AI, blockchain and Web3, Internet of Things (IoT), distributed systems, cloud and edge computing.

PROFESSIONAL EXPERIENCE **Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences** Shenzhen, China
Assistant Researcher (equiv. Assistant Professor) Jul. 2026 – Current

- Research on blockchain and artificial intelligence.
- Application for competitive research grants and projects.
- Publication of academic papers.

Postdoctoral Research Fellow Nov. 2023 – Jun. 2026

- Independent and collaborative research on decentralized AI, autonomous AI agents, AI governance, edge computing, and trustworthy distributed intelligence.
- Prototype development, system implementation, and experimental validation across distributed intelligent systems and networked computing infrastructures.

Southern University of Science and Technology Shenzhen, China
Research and Teaching Assistant Jul. 2021 – Sep. 2023

- Project *Distributed Resource Management for the Integrated Blockchain-Mobile Edge Computing System*.
- Academic research and teaching activities.

Ericsson Information Communication Ltd. Xi'an, China
Network Engineer (Internship) Feb. 2015 – Mar. 2015

- Radio Access Network design and optimization.

EDUCATION **Beijing Institute of Technology** Beijing, China
Ph.D. in Computer Science and Technology Sep. 2016 – Jun. 2023
Thesis: *Research on QoS-aware Software-Defined Information-Centric Networking Solutions*
Advisor: Zhu Liehuang

Northwestern Polytechnical University Xi'an, China
M.Eng. in Computer Science and Technology Sep. 2012 – Mar. 2015
Thesis: *Source Code-based Buffer Overflow Detection*
Advisor: Yao Ye

Northwestern Polytechnical University Xi'an, China
B.Eng. in Computer Science and Technology Sep. 2008 – Jul. 2012
Thesis: *Filtering packet from a sniffer type application on the IPv6 based networking environment*
Advisor: Jiang Xuefeng

PROJECTS

- **Role:** Co-PI; *AI-Enhanced Explainable Risk Intelligence Platform for Stablecoin Compliance*; Hangzhou StablePay Technology Co., Ltd.; **Amount:** RMB 120,000.
- **Role:** Technical Lead; *IT Consulting Services for JD Technology's Stablecoin Business*; JD Technology Co., Ltd.; **Amount:** RMB 350,000.

PATENTS

- *Task Processing Method, Apparatus, System, and Electronic Device*. Chinese invention patent, Application No. CN202510684562.0, 2025.

SELECTED PUBLICATIONS **Complete list:** Google Scholar · H-Index: 11 · i10-Index: 14 · Citations: 1229

1. **M. M. Karim**, Q. Qu, K. Sharif, M. Muzammal, and S. Biswas, "MTC-SBC: Reputation-based service provision for multi-tier computing-enabled sharded blockchain," *Future Generation Computer Systems*, vol. 182, p. 108499, Sep. 1, 2026. DOI, (JCR-Q1, IF=5.9).
2. **M. M. Karim**, S. Khan, Q. Qu, M. Muzammal, K. Sharif, and S. Biswas, "From trust to augmentation: A comprehensive survey on synergistic integration of decentralized and generative intelligence," *Computer Science Review*, vol. 61, p. 100936, Aug. 1, 2026. DOI, (JCR-Q1, IF=17.5).
3. **M. M. Karim**, D. Hoang Van, and Q. Qu, "Securing decentralized finance: A comprehensive survey of

maximal extractable value and its countermeasures,” *Blockchain: Research and Applications*, p. 100 455, Feb. 2, 2026. DOI, (JCR-Q1, IF=6.5).

4. **M. M. Karim**, Q. Qu, Y. Cai, T. Liu, and X. Meng, “Bitcoin reimagined: A comprehensive study of ordinals and inscriptions protocols for Web3 asset innovation,” *Blockchain: Research and Applications*, p. 100 379, Sep. 8, 2025. DOI, (JCR-Q1, IF=6.5).
5. **M. M. Karim**, S. Khan, D. H. Van, X. Liu, C. Wang, and Q. Qu, “Transforming data annotation with AI agents: A review of architectures, reasoning, applications, and impact,” *Future Internet*, vol. 17, no. 8, p. 353, Aug. 2025. DOI, (JCR-Q2, IF=4.6).
6. **M. M. Karim**, D. H. Van, S. Khan, Q. Qu, and Y. Kholodov, “AI agents meet blockchain: A survey on secure and scalable collaboration for multi-agents,” *Future Internet*, vol. 17, no. 2, p. 57, Feb. 2025. DOI, (JCR-Q2, IF=4.6).
7. **M. M. Karim**, K. Sharif, S. Biswas, Z. Latif, Q. Qu, and F. Li, “CIC-SIoT: Clean-slate information-centric software-defined content discovery and distribution for Internet of Things,” *IEEE Internet of Things Journal*, vol. 11, no. 22, pp. 37 140–37 153, Nov. 2024. DOI, (JCR-Q1, IF=8.7).
8. D. H. Van, **M. M. Karim**, and Q. Qu, “Just-in-time historical state reconstruction for low-latency financial trading with large language models,” *AI*, vol. 7, no. 4, p. 117, Apr. 2026. DOI, (JCR-Q1, IF=6.5).
9. S. Khan, **M. M. Karim**, Q. Qu, and M. Muzammal, “Dynamic client selector: A double deep q-learning framework for efficient federated learning,” in *Proceedings of the IEEE International Conference on Pervasive Computing and Communications*, Pisa, Italy, 2026, pp. 1–10. DOI.
10. D. H. Van, **M. M. Karim**, and Q. Qu, “Probabilistic optimization of top-k aggregation in distributed environments,” in *Web and Big Data*, Springer Nature, 2026, pp. 566–580. DOI.
11. L. Zhu, **M. M. Karim**, K. Sharif, C. Xu, and F. Li, “Traffic flow optimization for UAVs in multi-layer information-centric software-defined FANET,” *IEEE Transactions on Vehicular Technology*, vol. 72, no. 2, pp. 2453–2467, Feb. 2023. DOI, (JCR-Q1, IF=7.5).
12. L. Zhu, **M. M. Karim**, K. Sharif, C. Xu, F. Li, X. Du, and M. Guizani, “SDN controllers: A comprehensive analysis and performance evaluation study,” *ACM Computing Surveys*, vol. 53, no. 6, 133:1–133:40, Dec. 6, 2020. DOI, (JCR-Q1, IF=30.4).
13. W. Ni, S. Zhu, **M. M. Karim**, A. Asheralieva, J. Kang, Z. Xiong, and C. Maple, “Lagrange coded federated learning (L-CoFL) model for Internet of Vehicles,” in *2022 IEEE 42nd International Conference on Distributed Computing Systems*, Jul. 2022, pp. 864–872. DOI.
14. S. Zhu, A. Asheralieva, **M. M. Karim**, D. Niyato, and K. A. Raza, “Secure federated learning based on coded distributed computing,” in *2021 IEEE Globecom Workshops*, Dec. 2021, pp. 1–7. DOI.
15. Z. Latif, K. Sharif, F. Li, **M. M. Karim**, S. Biswas, M. Shahzad, and S. P. Mohanty, “DOLPHIN: Dynamically optimized and load balanced path for inter-domain SDN communication,” *IEEE Transactions on Network and Service Management*, vol. 18, no. 1, pp. 331–346, Mar. 2021. DOI, (JCR-Q1, IF=5.7).
16. Z. Latif, K. Sharif, F. Li, **M. M. Karim**, S. Biswas, and Y. Wang, “A comprehensive survey of interface protocols for software defined networks,” *Journal of Network and Computer Applications*, vol. 156, p. 102 563, Apr. 15, 2020. DOI, (JCR-Q1, IF=7.7).
17. M. W. A. Ashraf, C. Huang, A. Raza, K. Sharif, **M. M. Karim**, and S. Huang, “Forwarding and caching in video streaming over ICSDN: A clean-slate publish-subscribe approach,” *Computer Networks*, vol. 219, p. 109 433, Dec. 24, 2022. DOI, (JCR-Q1, IF=4.7).
18. I. Alam, K. Sharif, F. Li, Z. Latif, **M. M. Karim**, S. Biswas, B. Nour, and Y. Wang, “A survey of network virtualization techniques for Internet of Things using SDN and NFV,” *ACM Computing Surveys*, vol. 53, no. 2, 35:1–35:40, Apr. 16, 2020. DOI, (JCR-Q1, IF=30.4).
19. A. Bilal, K. Sharif, L. Zhu, F. Li, C. Xu, and **M. M. Karim**, “HySLA: Hybrid DPoS-DAG model for secure, scalable, and low-latency access control in Internet of Vehicles,” *IEEE Internet of Things Journal*, vol. 13, no. 9, pp. 18 930–18 944, May 1, 2026. DOI, (JCR-Q1, IF=8.7).
20. A. Bilal, K. Sharif, L. Zhu, F. Li, C. Xu, and **M. M. Karim**, “Evaluation to integration: Hybrid feature selection framework with ensemble machine learning for intrusion detection,” *IEEE Transactions on Dependable and Secure Computing*, vol. 23, no. 3, pp. 6362–6377, May 2026. DOI, (JCR-Q1, IF=6.8).

ACADEMIC SERVICE

Conference Organization

- Award Chair, IEEE International Conference on Big Data Science and Engineering
- Technical Program Committee Member, International Conference on Communication Systems and Networks

Peer Review

- All the leading journals and conferences of IEEE, ACM, Elsevier, and Springer.